Internship Project Completion Report

Digital Transformation Initiatives in Maintenance & Business Operations

Factory Floor Maintenance Dashboard
Vendor Evaluation & Performance Management System (VEMS)
MPAS Store Digitisation

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Overview

This report consolidates three digital transformation initiatives undertaken across two departments during a summer internship: the **Factory Floor Maintenance Dashboard**, built while working in the **Maintenance department**, a maintenance tool to reduce equipment downtime on the shop floor; and the **Vendor Evaluation and Performance Management System (VEMS)** and the **MPAS Store Digitisation** project, both undertaken while working in the **Operations and Business Excellence (OS&BE) department**.

Each initiative targets a different point in the same operational chain. The Dashboard addresses maintenance response time on the shop floor once equipment is already running. MPAS Store Digitisation addresses how material is received, stored, and issued before it ever reaches the floor. VEMS addresses the vendor relationships further upstream, where delivery and quality issues originate before they ever become a store or floor problem. Together they represent a connected view of the plant's operations, from vendor performance through material handling to floor-level execution.

This report is structured in three parts, one per project, each covering the problem it addresses, the solution designed, the work completed, and what the project contributed to the broader internship learning.

Part I

Factory Floor Maintenance Dashboard

1 Dashboard: Executive Summary

This part covers the design, development, and deployment of the **Factory Floor Maintenance Dashboard**: a live web application that digitises equipment status logging, asset tracking, material code tool readiness lookup, and cross-shift communication on the shop floor.

This is a maintenance tool, built during the portion of the internship spent in the plant's **Maintenance department**, as part of a broader digital transformation effort. The system

replaces a set of disconnected manual routines with a single shared platform that every operator, technician, and shift can use simultaneously.

The sections below walk through the problem it solves, how it was built, how it is expected to reduce maintenance downtime, and what the project taught about both software engineering and working inside a real production environment.

2 Dashboard: What We Were Trying to Solve

Before this system, maintenance and production information on the floor moved through four separate and largely disconnected channels.

- Paper checklists for equipment status, filled in once per shift and rarely seen by anyone outside that shift.
- Verbal escalation of faults: an operator had to physically walk to maintenance or call someone to report a problem.
- Static Excel sheets for asset registers and spare-part mappings, often out of date and slow to search under time pressure.
- Tribal knowledge: how specific machines fail and how to fix them, held in individual technicians' heads rather than written down anywhere.

None of these talked to each other. A fault could be resolved by one shift and re-reported by the next simply because no one had a shared record. Tool availability for the next shift depended on whoever happened to notice a gap, with no structured way to verify readiness against the production plan.

3 Dashboard: Project Goals

The dashboard was built around five clear objectives.

1. Replace paper-based equipment checks with a structured digital logging form.
2. Give every shift a single, live view of every open fault on the floor.
3. Encode common troubleshooting steps into a guided assistant so response quality does not depend on who is on shift.
4. Centralise the asset registry into one searchable system covering tools, controllers, IPs, locations, and reference documents.
5. Surface the right tools and their stage locations and required counts from a material code lookup, so readiness can be verified before a shift begins rather than discovered as a gap when the line needs them.

4 Dashboard: What the System Does

The application is a single-page React/TypeScript app backed by a Firebase Realtime Database, meaning every change one user makes is visible to everyone else instantly, with no refresh needed.

4.1 Role-Based Login

Users sign in as an Operator or a Technician, selecting their shift (A, B, or C) and the date. Technicians have a password gate and additional capabilities: resolving incidents, modifying assets, and resetting data. Operators see everything but cannot alter records.

4.2 Live Dashboard

The home screen shows what a shift lead needs at a glance: total active alarms, total registered assets, a faults-by-category chart (Tools, Panels, UPS, TVs/Scanners), and the open incidents wall showing every unresolved fault with its location, IP address, and reported problem.

4.3 Five-Step Equipment Logging Form

Instead of a blank paper form, operators work through a short guided sequence: equipment details, a category-specific checklist (temperature and MCB status for panels; current load and filter status for UPS), an overall status selection, and if something is wrong, root cause, action taken, and whether the ticket should stay open. Submitting triggers an automatic alert if the issue is unresolved.

4.4 Team Intercom and SOP Assistant

A live chat channel keeps everyone in the same conversation, with image attachments for visual problems. System messages such as fault alerts and resolution confirmations appear alongside manual messages, so there is one complete timeline for each shift.

Inside the chat is the SOP Assistant: a guided troubleshooting flow covering torque wrenches, UPS units, barcode scanners, Andon displays, Moxa serial-to-Ethernet devices, and machine-vision systems. An operator picks the machine and the symptom, and the system walks them through the fix. If the problem is not on the list, it is escalated as a new ticket directly to a technician.

4.5 Asset Registry

Every tool and device is registered with its make, controller serial, network IP, and stage location. Assets can be added one at a time or imported in bulk from Excel, with unrecognised columns preserved as custom fields.

4.6 Material Code Tool Readiness Lookup

This is the feature most directly aimed at reducing MTTR before a fault ever happens, by changing tool readiness from something discovered reactively to something verified proactively.

The problem it replaces: every day's production plan calls for a specific set of tools at specific stages, in specific quantities. Previously there was no way to see this picture in advance. A technician only found out a torque wrench was missing, miscalibrated, or short on count when a line operator was already stuck waiting for it. The check happened at the point of failure, not before it.

How it works: at the start of a shift, a technician enters the material codes corresponding to that day's production plan into the lookup. The system instantly returns every tool linked to those material codes, along with two pieces of information for each one: the **stage ID** where it is required, and the **count** needed at that stage. Instead of a flat list of "tools in inventory," the technician sees exactly what is needed, where, and how many, mapped directly to what the floor will actually run that day.

What this enables: with that list in hand before the shift starts, the maintenance team can walk the floor and physically confirm each tool is present at its stage, in working condition, and

in the required count, and flag or replace anything missing before the line ever asks for it. This moves the readiness check from a reactive scramble during production to a five to ten minute proactive sweep at the start of the shift.

Why this reduces MTTR specifically: a large share of maintenance downtime is not caused by a part breaking, it is caused by the right tool not being where it is needed when it is needed. By surfacing the full day's tool requirement up front, the system removes the most preventable category of delay from the MTTR chain entirely, before detection or diagnosis even comes into play.

4.7 Shift Log History

Every status check and fault is recorded with a timestamp, operator name, shift, and outcome. The log is searchable, filterable by category, and exportable to CSV, turning routine check-ins into a dataset that can support root-cause analysis over time.

5 Dashboard: How It Was Built

- **Frontend:** React 18 with TypeScript, built using Vite, styled with Tailwind CSS for a clean, shop-floor-friendly interface that works on both desktop terminals and mobile devices.
- **Database:** Firebase Realtime Database, chosen because it pushes updates to every connected client instantly, no polling needed.
- **Excel Import:** SheetJS parses existing asset registers and spare-parts sheets directly in the browser, auto-mapping recognised column names and preserving unrecognised ones as custom fields.
- **Access Control:** A two-role model (Operator and Technician) gates sensitive actions without the overhead of a full identity management backend.
- **Deployment:** Packaged with Vite and deployed on Vercel, giving every shift a single always-current URL with no installation.

6 Dashboard: Impact on Maintenance Response Time (MTTR)

Mean Time to Repair (MTTR) is the average time between a piece of equipment failing and being restored to normal. It is not one activity but the sum of several sequential delays, and every one of those delays is shaped by how information moves through the plant.



Phase	Traditional Bottleneck	What the Dashboard Does
Detection	A fault is only known once someone notices it and chooses to report it verbally.	The logging form and incident wall make a fault visible to the whole floor the moment it is recorded.
Notification	Someone must physically find or call a technician.	Logging a fault automatically posts a tagged alert into the team chat. Technicians see it the moment it is created.
Diagnosis	Quality varies by which technician is on shift and what they happen to remember.	The SOP Assistant gives every operator the same documented steps for known fault patterns, and escalates unfamiliar ones directly.
Logistics	Tool availability was checked only when a line operator was already stuck waiting.	Entering the day's material codes shows every required tool, its stage, and count up front, so the team confirms readiness before the shift, not after a stoppage.
Repair and Verification	No reliable record of whether a fix actually happened or was communicated back.	One-click resolution marks the log as fixed, resets the asset status, and posts a confirmation visible to the whole floor.

6.1 Benefits That Build Over Time

The system also produces gains that compound as it gets used.

- Incoming shifts start with full context instead of relying on a verbal handover that might miss details.
- Guided troubleshooting narrows the gap between new and experienced staff, so response quality is more consistent across the team.
- The shift log accumulates into a searchable record of faults, making it possible to spot recurring failure patterns on specific assets rather than treating each breakdown as a one-off.
- Automatic alerts and resolution confirmations prevent two people from independently investigating the same issue.

After adoption, the main remaining component of MTTR is the physical repair itself, which is the one part a digital system cannot and should not try to shorten.

7 Dashboard: What I Learned

7.1 Software and Engineering

- **Full-stack development:** Building a multi-view application with shared live state, typed interfaces, form validation, and reusable components from scratch in React and TypeScript.
- **Real-time data:** Designing around Firebase Realtime Database so that every user sees consistent state without manual refreshes or background polling.
- **Legacy data migration:** Parsing existing Excel-based registers and BOM sheets directly in the browser with SheetJS, automatically mapping column names and preserving unrecognised

fields rather than discarding them, which is what made the system actually usable by people who already had their data in spreadsheets.

- **Role-based design:** Building just enough access control to protect sensitive actions without adding complexity that would slow down everyday operators.
- **Designing for the floor:** The system is used standing at a terminal between tasks, not sitting at a desk. That meant larger touch targets, minimal free-text entry, and clear colour coding for status at a glance.
- **Build and deployment:** Packaging with Vite and deploying on Vercel so any terminal with a browser can use the system without installation or IT involvement.

7.2 Working Inside a Real Plant

- **Reactive vs. preventive vs. predictive maintenance.** Most floor activity sits somewhere between fixing things when they break and servicing them on a schedule. Consistent digital logging is the foundation that makes a predictive approach possible: without data, there is nothing to analyse for early warning signs.
- **MTTR and MTBF as real language.** These are not abstract numbers from a textbook. They are how a team describes whether a line is healthy, and every minute of MTTR traces back to a specific hand-off that either worked or did not.
- **SOPs as organisational memory.** A troubleshooting procedure is only useful if it is easy to find exactly when it is needed. Moving SOPs from binders to an in-context tool changes who can respond effectively, not just how fast.
- **Calibrated tools are not just inventory.** Torque wrenches carry controller serials, port mappings, and calibration status as part of quality traceability. Treating them as inventory items without that context loses critical information.
- **Tool availability is a maintenance problem.** Knowing a tool exists somewhere on the floor is not the same as knowing it is at the right stage, in the right count, and ready for the next shift. Linking material codes to stage locations and required counts makes that readiness verifiable rather than assumed.
- **Shift handover is a vulnerability.** A huge amount of floor knowledge travels verbally between shifts and gets lost or distorted. Systems that share state rather than hand it over reduce dependence on any one conversation going perfectly.
- **Adoption matters more than completeness.** A simpler form that operators will fill in every time is worth more than a comprehensive one that gets skipped under time pressure. Designing for the floor, not for the spec sheet, was the clearest lesson of the internship.

8 Dashboard: Conclusion and What Comes Next

The dashboard brings five previously disconnected workflows: status logging, fault escalation, troubleshooting, asset tracking, and material code tool readiness lookup, into one live system. By closing the information gaps between these steps, it targets every phase of MTTR except the physical repair itself, and produces a growing record that supports longer-term reliability work.

Possible next steps include:

- Analytics on the accumulated shift-log data to surface recurring failure patterns by asset or production line.
- Calibration due-date tracking so the asset registry can flag tools approaching their recalibration window automatically.
- Push notifications for critical alerts, so technicians do not need to have the intercom actively open to catch new faults.

Part II

Vendor Evaluation & Performance Management System (VEMS)

9 VEMS: Problem Statement

Vendor performance at a manufacturing plant is tracked, if at all, through a combination of informal feedback, periodic meetings, and spreadsheets maintained by individual buyers. When a vendor repeatedly delivers late, ships wrong parts, or sends material that fails SQA inspection, there is no single place that holds that history in a structured, comparable form. Decisions about whether to continue with a vendor, renegotiate terms, or escalate to procurement leadership are made largely on institutional memory rather than on a consistent, auditable record.

This creates three compounding problems:

- **No vendor accountability baseline.** Without a documented performance score, conversations about a vendor's reliability are opinion-based. Poor performers face no structured consequence and strong performers receive no recognition.
- **No trend visibility.** A vendor who was reliable for two years and has been deteriorating for the last three months looks identical in the absence of historical tracking.
- **Audit exposure.** If a quality escape reaches the field and traces back to a supplier, the absence of documented incoming inspection outcomes and vendor scores is a significant liability.

VEMS was initiated to replace this fragmented picture with a structured digital system that scores every active vendor against a defined KPI framework, tracks performance over time, and gives procurement and quality teams a single point of reference for all vendor-related decisions.

10 VEMS: Project Goals

1. Define a structured, weighted KPI framework that covers all dimensions of vendor performance relevant to the plant.
2. Produce a technical BRD that specifies the system in enough detail for a development team to build it without further requirements gathering.
3. Design a UI concept that makes vendor scorecards fast to read and act on, distinct from the plant's existing SAP interface.

4. Specify an infrastructure approach that keeps all vendor data fully on-premises, satisfying the data confidentiality constraint without requiring cloud exemptions.

11 VEMS: KPI Framework

The evaluation framework was developed from an existing confidential vendor assessment workbook and structured into **26 sub-KPIs across five performance categories**. Each sub-KPI has a defined measurement method, data source, scoring scale, and green/amber/red threshold. The composite vendor score is a weighted aggregate across all five categories.

Category	Sub-KPIs Covered
Delivery Performance	On-time delivery rate, lead time adherence, advance shipment notice (ASN) compliance, dock-to-stock cycle time, and emergency delivery frequency.
Quality Performance	Incoming inspection pass rate, parts-per-million (PPM) defect rate, SQA rejection frequency, field return rate, and repeat defect recurrence.
Responsiveness & Communication	Speed of response to ASN exceptions, ODC resolution turnaround, buyer query response time, escalation handling, and proactive issue communication.
Documentation & Compliance	Invoice accuracy rate, GRN supporting document completeness, regulatory and certification compliance, and audit readiness score.
Commercial & Relationship	Price stability, responsiveness to process improvement requests, dispute resolution behaviour, and strategic partnership contribution.

Vendors are scored on each sub-KPI on a 1–5 scale. Category weights are configurable by the procurement team to reflect the plant's current strategic priorities, allowing the composite score to be recalibrated without changing the underlying data. A vendor detail view shows the composite score, a category-level breakdown, and a sub-KPI drill-down so the source of any red flag is always traceable.

12 VEMS: System Design and Infrastructure

12.1 The Self-Hosted Constraint

The most consequential design decision for VEMS was made before a single line of code was written: vendor performance data cannot leave the plant's network. Scores, evaluation histories, and the supplier names attached to them are commercially sensitive. This ruled out all external cloud hosting options and shaped every subsequent infrastructure choice.

VEMS is designed to run entirely on-premises on a Windows Server instance already managed by the plant's IT team, using only software that can be maintained without cloud-specific expertise or ongoing vendor subscriptions.

12.2 Proposed Stack

- **Web server:** Internet Information Services (IIS) serves the frontend build and acts as a reverse proxy routing API requests to the application layer.

- **Application server:** Node.js managed under PM2. PM2 handles process persistence, automatic restart on failure, log rotation, and startup on server reboot without additional configuration.
- **Database:** PostgreSQL on Windows Server. Chosen for its reliability with relational, structured data, strong support for the KPI scoring schema and weighted aggregate queries, and zero licensing cost, which matters for an internal tool with no budget line of its own.

12.3 UI Concept

The frontend evaluation interface was designed using a glassmorphism visual language: frosted-glass card surfaces, translucent panels, layered depth, and a dark base palette. The decision was deliberate: evaluators using VEMS should immediately know they are in a tool designed for vendor review, not another SAP screen.

The vendor list view allows filtering by category score, composite score range, and evaluation cycle. The vendor detail view surfaces the composite score prominently at the top, with category-level performance rendered as a radial chart so the shape of a vendor's strengths and weaknesses is readable in seconds. Each category card expands to show the sub-KPI breakdown with colour-coded status indicators. Historical trend data is shown as a sparkline on the vendor card, making performance drift visible without opening the detail view.

13 VEMS: BRD and Deliverables

The primary output from the VEMS workstream during this internship is a series of iterations of the technical **Business Requirements Document (BRD)**, authored in LaTeX and revised through multiple review cycles with the Operations and Business Excellence team.

What the BRD Covers

- Problem statement and current-state gap analysis for vendor evaluation.
- Functional requirements for the KPI scoring engine, vendor profile management, evaluation cycle management, historical trend views, and report export.
- Non-functional requirements: data residency (on-premises only), role-based access control, audit logging of all score changes, backup and recovery policy, and performance targets.
- Data model: entity definitions for vendors, evaluation cycles, sub-KPI scores, category weights, and weighted composite score calculation.
- Infrastructure specification: Windows Server deployment procedure, IIS reverse proxy configuration, Node.js PM2 process management setup, and PostgreSQL schema design.
- UI wireframe descriptions aligned to the glassmorphism design concept, covering vendor list, vendor detail, evaluation form, and trend views.

What Comes Next

The BRD is the primary handover artefact for the development team that will implement VEMS. It was written to be actionable: a developer picking it up should be able to begin without a requirements discovery session. VEMS remains in the requirements and design phase at the close of this internship.

14 VEMS: What I Learned

- **A KPI framework is only as good as its measurement method.** Defining 26 sub-KPIs is straightforward. Defining how each one is measured, from what data source, at what frequency, and by whom, is where the real work is. Several early sub-KPI definitions had to be reworked because the data they required did not exist in a form the system could consume.
- **Confidentiality shapes architecture before features do.** The self-hosted constraint was not a technical preference; it was a business requirement. Starting from that constraint and working outward to the stack is a different process than picking the best cloud option and then worrying about data governance.
- **Writing a BRD for a developer is different from writing a requirements document for a stakeholder.** The stakeholder needs to agree with what the system should do. The developer needs to know exactly what to build. The revision cycles on the BRD were largely about closing the gap between those two audiences.
- **UI design for internal tools still matters.** An interface that evaluators find tedious to use will produce low-quality, rushed scores. The glassmorphism approach was not aesthetic preference; it was an attempt to make the evaluation task feel like it deserved attention.

Part III

MPAS Store Digitisation

15 MPAS: Problem Statement

The MPAS (Material Planning and Store) process covers every step from a vendor vehicle arriving at the plant gate to a picked part reaching the assembly line. At the time of this analysis, the process was supported by a combination of SAP for formal transactions, a set of offline Excel files maintained by individuals, and handwritten registers, three of which exist in parallel for every single inbound shipment, none of them linked to the others.

The consequences are visible on the floor every shift:

- **GRN lag.** Material arrives and is physically placed in storage, but SAP stock is not updated until the DEO processes all GRNs together at the end of the shift. Planners work on stale data for up to eight hours, generating false shortages and unnecessary escalations.
- **Wrong-bin placements.** Bin locations are looked up in an offline Excel file shared via WhatsApp. The file is accurate only at the moment of download. Placements made against an outdated file fail at the point of picking, often a full shift later.
- **Line stoppages from wrong-part issuance.** Picking is done by reading a printed list and physically selecting parts by eye. There is no scan at any point of pick. Wrong part, wrong revision, and wrong quantity are undetectable until the assembler encounters the mismatch.
- **Lost ODC records.** Over, Damage, and Shortage events are noted by hand on the physical invoice. If the invoice is misplaced, the ODC record is permanently lost with no photo, no timestamp, and no system flag.

16 MPAS: Project Goals

1. Map every step of the end-to-end store process from gate to line in a single consolidated document, resolving inconsistencies across the four source process sheets.
2. Assess every step for process type (manual, semi-digital, SAP-driven), identify what specifically goes wrong and why, and rate the consequence.
3. Identify the systemic root causes behind the pattern of high-risk steps rather than treating each issue as isolated.
4. Produce a prioritised improvement roadmap that a development team can use as the basis for the MPAS digitisation BRD.

17 MPAS: Current-State Analysis

The analysis covers **50 process steps across five phases**, consolidated from four source documents: the Bill of Flow (BOF, 29 steps), Store Process Flow (SPF, 9 steps), Store Material Binning Sheet (6 steps), and Work Order and Line Feeding Sheet (11 steps). Where BOF and SPF described the same physical action, steps were merged; where SPF introduced information absent from BOF, it was added.

17.1 Process Type Legend

Symbol	Type	Meaning
●	Manual / Handwritten	Fully human-dependent. No digital tool. Highest error and audit risk.
●	Semi-Digital	A digital tool is used but disconnected, stale, or uncontrolled. Medium risk.
●	SAP-Driven	SAP is the system of record. Risk depends on whether entry is manual or scan-triggered.

17.2 Risk Distribution Across Phases

Phase	Steps	Critical (9–10)	High (7–8)	Medium (5–6)
A: Gate Entry & Documentation	11	1	6	3
B: Unloading, Inspection & Receiving	13	5	7	1
C: GRN Processing & Closure	9	3	4	2
D: Store Binning / Putaway	6	3	3	0
E: Work Order Picking & Issuance	11	5	5	1
Total	**50**	**17**	**25**	**7**

Step 16 (manual count at unloading) and Step 45 (manual work order picking) were both rated 10/10, the highest risk points in the entire process.

17.3 Four Root Causes Behind the Numbers

1. **Three disconnected registers per shipment.** Security, receiver, and POD each maintain a separate handwritten register for the same inbound lot. None link to SAP; none are recon-

ciled. Every audit query requires manual cross-referencing across all three, and discrepancies between them cannot be resolved objectively.

2. **Offline MROBIN Excel as the bin location master.** The MROBIN file is a SAP snapshot shared via WhatsApp to team members' phones. Multiple versions circulate simultaneously. Any bin update made in SAP after the last download is invisible. This gap is the root cause of most wrong-bin placements and the picking failures they produce.

3. **Batch GRN entry at shift end.** Material arrives and is moved to storage, but SAP stock is not updated until the DEO batch-processes all GRNs at shift end, up to eight hours later. During this window, planners see incorrect stock and make procurement decisions on stale data.

4. **No scan at any point of pick or putaway.** From unloading through binning through picking, not one step uses a scan to validate the right part at the right location. All validation is visual or manual, making errors undetectable until they surface at the line.

18 MPAS: Phase-by-Phase Summary

18.1 Phase A: Gate Entry and Documentation (Steps 1–11)

The process begins with no pre-arrival visibility: security has no ASN-based expected vehicle list, and the receiving team receives no system notification that a shipment is incoming. All gate data entry into SAP (T-code ZMGAE) is fully manual. Typos at this step propagate downstream into the GRN, inventory record, and Material Identification Tag without any validation catch. ASN mismatches trigger a dual-channel escalation via email and WhatsApp with no SLA; vehicles routinely wait hours at the dock during resolution.

Highest risk: Step 2 (handwritten security register), rated ***CRITICAL 9/10***.

18.2 Phase B: Unloading, Inspection and Receiving (Steps 12–24)

This is the highest-risk phase. Part number and quantity verification (Step 16) is entirely manual: visual count, no scan, no system validation against the open PO. Wrong parts and wrong quantities accepted here are undetectable until the line encounters the mismatch. SQA inspection (Step 18) runs as an offline parallel process; approval travels by WhatsApp with no system hold status, meaning rejected material has been physically placed in storage bins when the communication was missed. The phase involves seven separate physical documents or register entries for a single shipment, none integrated with SAP.

Highest risks: Steps 16 and 18, both rated ***CRITICAL 9–10/10***.

18.3 Phase C: GRN Processing and Documentation Closure (Steps 25–33)

GRN is batch-processed at shift end rather than at the point of physical receipt, creating up to an eight-hour gap between physical and system stock. ODC tracking lives entirely outside SAP in a personal Excel file and email threads with no SLA, no single owner, and no escalation path. The hold on an invoice pending ODC resolution is informal, not a system flag, meaning DEO must maintain a manual mental list of which invoices are on hold.

Highest risk: Step 29 (batch GRN at shift end), rated ***CRITICAL 9/10***.

18.4 Phase D: Store Binning and Putaway (Steps 34–39)

Three of the six steps in this phase are rated Critical. The MROBIN file drives all three: wrong-bin placements from an outdated file (Steps 35–36), and a 1–3 day email chain for new part

locations that leaves parts invisible to pickers during the resolution window (Step 37). There is no scan at the destination bin at any point; correct placement is entirely dependent on the operator reading the bin code correctly.

Steps 35, 36, and 37 all rated ***CRITICAL 9/10***.

18.5 Phase E: Work Order Picking and Line Issuance (Steps 40–50)

The production plan arrives over two unofficial channels (email and WhatsApp) that must be manually reconciled each morning. The picklist is downloaded from SAP as a static Excel file, manually recreated as a printed issue list, and physically distributed area by area before picking starts, by which point it may already be a superseded version of the plan. Picking itself (Step 45) uses no scan; the operator reads a printed sheet and selects parts by eye. Shortage declaration (Step 48) is reactive: the email goes out after the line is already starved. Phone-based shortage follow-up (Step 49) has no record, no SLA, and no management visibility.

Highest risk: Step 45 (manual picking, no scan), rated ***CRITICAL 10/10***.

19 MPAS: Improvement Roadmap

19.1 Priority 1: Highest Consequence, Address First

The 30 Priority 1 items address all four root causes and target every step rated Critical or High with an upstream dependency on infrastructure that must exist first. The following are the most consequential:

- **Scan-based receiving at unloading (Step 16).** Operator scans part barcode against the open PO in SAP. System validates part number, quantity, and revision before the vehicle departs. This is the single highest-value change in the process.
- **Live bin lookup replacing MROBIN Excel (Steps 13, 35, 36).** MROBIN Excel eliminated entirely. Bin lookup is a live SAP query via MPAS mobile app: scan part QR, receive current bin in real time. No download, no stale file, no version divergence.
- **Real-time GRN at point of receipt (Step 29).** GRN posted in SAP at the moment of physical receipt. Scan-based receiving at Step 16 can trigger GRN automatically, collapsing the batch-processing lag.
- **Scan-guided picking (Step 45).** MPAS shows the pick task, operator scans the part QR, system validates part number, revision, and quantity before the operator moves the material. Mismatch is blocked, not noted after the fact.
- **SAP-based ODC management (Steps 21, 27, 28).** ODC raised directly in SAP at unloading with mandatory photo. System auto-notifies buyer, applies a formal PO line block, and auto-escalates if unresolved after a defined SLA window.
- **New part location SLA (Step 37).** Unrecognised part scan auto-creates a SAP task with a 4-hour SLA and assigns an interim location immediately, eliminating the 1–3 day email chain.

19.2 Five Steps That Disappear on Digitisation

A specific class of steps exists only as compensating workarounds for the lack of system integration at earlier points. When the upstream step is digitised, these steps are automatically eliminated with no separate effort:

Step	Why It Exists Now	What Eliminates It
14	Bin location recorded a second time after Step 13 lookup	Live bin lookup at Step 13 removes the need for any register entry
24	Third POD register maintained separately	POD digitised in Step 23; SAP holds all records
25	ODC paper form physically walked to DEO	ODC raised in SAP at Step 21; DEO sees it in system queue
43	Issue list manually recreated from picklist Excel	Live MPAS picklist replaces the downloaded Excel
44	Issue list physically distributed area by area	Operators receive pick tasks on mobile; no distribution needed

19.3 Priority 2 and 3

Priority 2 covers the 15 High-rated steps that depend on Priority 1 infrastructure being in place first, including FIFO batch management through SAP, digital POD generation, scan-confirmed material handover, and line feeding confirmation via mobile scan.

Priority 3 covers the 7 Medium-rated steps representing process optimisation once the core digitisation is stable: digital gate pass generation, real-time dock management, digital invoice archiving in SAP DMS, and real-time gate-out at dock clearance.

20 MPAS: What I Learned

- **The same information recorded three times is not three times safer.** Three disconnected registers for one shipment create three opportunities for error and three records that must be manually reconciled when they disagree. Redundancy without linkage is noise, not control.
- **An offline file is not a digital system.** The MROBIN Excel shared via WhatsApp has all the failure modes of a paper register: it gets out of date, diverges across copies, and cannot be searched reliably under pressure, plus the additional risk that it can be accidentally edited. The distinction between a digital tool and a system-integrated tool was the most important conceptual distinction in the analysis.
- **Batch processing is invisible debt.** GRN entry at shift end looks efficient because all the SAP transactions happen in one sitting. In reality, for the hours before that sitting, every decision by a planner, buyer, or store operator is made on wrong data. The cost is invisible in the transaction log but very visible in shortage calls and emergency procurement.
- **A missing scan is a missing gate.** The entire picking and receiving process has no scan at any point of action. Errors surface only when their consequences arrive at the line. Moving the catch point from assembly to picking and receiving is the single highest-value structural change available.
- **Process mapping requires listening more than observing.** Written procedures and actual practice diverged significantly in several areas. The most important issues, like SQA approval travelling by WhatsApp with no system integration, were not visible in any document. They emerged only from conversations with the people doing the work.

Closing Summary

The three initiatives in this report cover three different points in the same operational chain. The Factory Floor Maintenance Dashboard targets MTTR once equipment is already running on the line. MPAS Store Digitisation targets how material is received, stored, and issued before it reaches that line. VEMS targets the vendor relationships further upstream, where many of the issues that surface on the floor and in the store, including late deliveries, wrong parts, and quality escapes, originate in the first place.

The Dashboard is live and deployed. The MPAS and VEMS workstreams close this internship with a 50-step current-state process map and risk-rated improvement roadmap, and a fully iterated technical BRD respectively, both serving as the documented foundation for the next phase of development.